MEASUREMENT

Distance-Time Graphs

Name _____ Date _____ No calculator Version 1

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KEY WORDS distance-time graph · speed *(tere)* · steep · stationary · journey · gradient *(rōnaki)*

A Reading a distance-time graph ●○○

Remember: Time goes along the bottom and distance from home goes up the side. A flat line means the person has stopped.

1. The graph shows a bike ride that Mia took from home.



a) How far from home is Mia after 1 hour? _____

b) What is the furthest that Mia gets from home? _____

c) For how long does Mia stop? _____

d) After how many hours does Mia get home? _____

e) How far does Mia ride altogether? _____

B Speed from a graph ●●○

Remember: Speed = distance ÷ time. On a distance-time graph, a steeper line means a faster speed.

2. Use the graph of Mia's bike ride above.

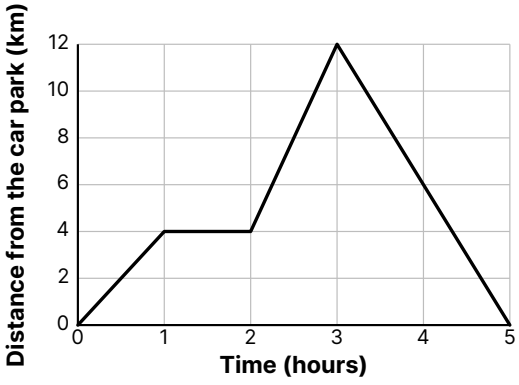
a) Find Mia's speed on the way out.

b) Find Mia's speed on the way home.

c) Was Mia faster on the way out or on the way home? How does the graph show this?

C Describing a journey ●●●

3. The graph shows a walk in the hills. Describe what is happening in each part of the walk, and give the speed when the walker is moving.



a) From 0 to 1 hour

b) From 1 to 2 hours

c) From 2 to 3 hours

d) From 3 to 5 hours

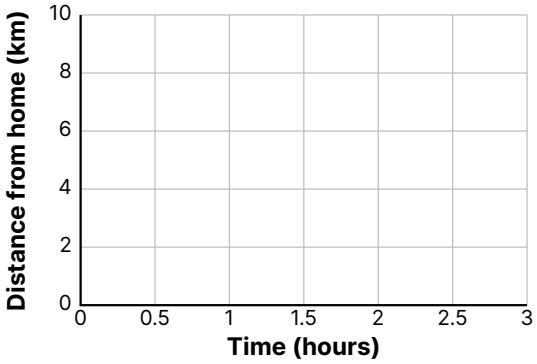
e) In which part is the walker moving fastest?

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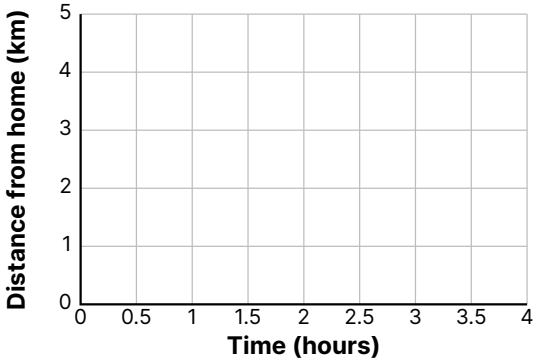
D Drawing a journey ●●●

Remember: Plot a point at the end of each part of the journey, then join the points with straight lines.

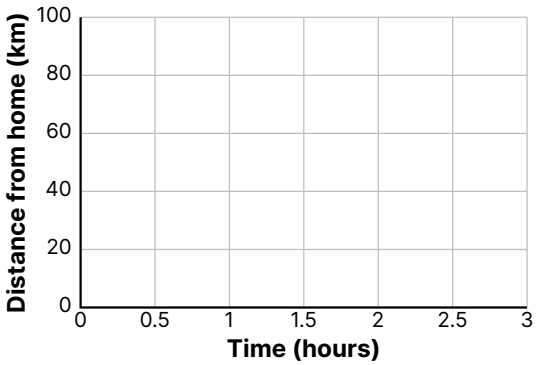
4. Hemi leaves home and cycles 8 km in 1 hour, then stops for 30 minutes, then cycles straight home, which takes 1 hour 30 minutes. Draw the journey on the grid.



5. Ana leaves home and walks 4 km in 1 hour, then stops for 1 hour, then walks straight home, which takes 1 hour 30 minutes. Draw the journey on the grid.



6. A truck driver leaves home and drives 60 km in 1 hour, then stops for 30 minutes, then drives 40 km further from home in 1 hour. Draw the journey on the grid.



Extension

7. Look again at the first graph on this sheet. Find the average speed for the whole ride, including the stop.

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Distance-Time Graphs

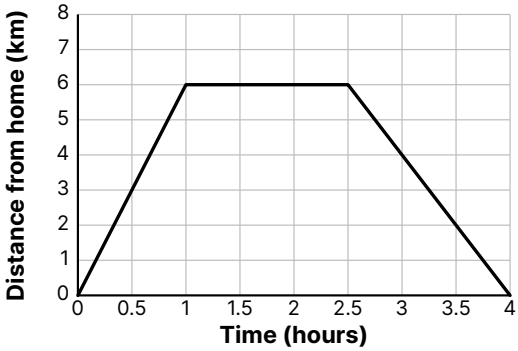
Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS distance-time graph · speed *(tere)* · steep · stationary · journey · gradient *(rōnaki)*

A Reading a distance-time graph ●○○

Remember: Time goes along the bottom and distance from home goes up the side. A flat line means the person has stopped.

1. The graph shows a bike ride that Tama took from home.



a) How far from home is Tama after 0.5 hours? _____

b) What is the furthest that Tama gets from home? _____

c) For how long does Tama stop? _____

d) After how many hours does Tama get home? _____

e) How far does Tama ride altogether? _____

B Speed from a graph ●●○

Remember: Speed = distance ÷ time. On a distance-time graph, a steeper line means a faster speed.

2. Use the graph of Tama's bike ride above.

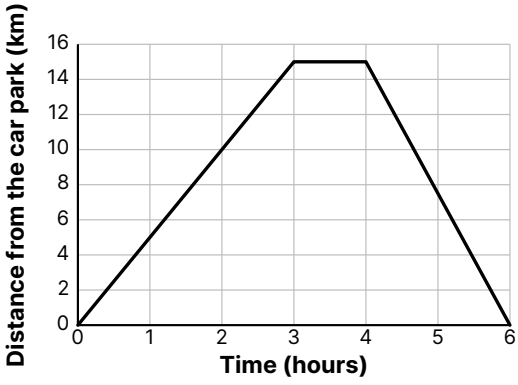
a) Find Tama's speed on the way out.

b) Find Tama's speed on the way home.

c) Was Tama faster on the way out or on the way home? How does the graph show this?

C Describing a journey ●●●

3. The graph shows a walk in the hills. Describe what is happening in each part of the walk, and give the speed when the walker is moving.



a) From 0 to 1 hour

b) From 1 to 3 hours

c) From 3 to 4 hours

d) From 4 to 6 hours

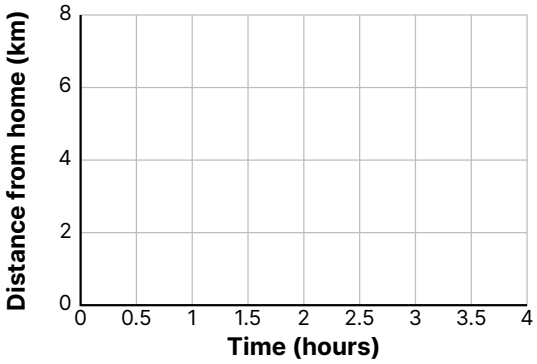
e) In which part is the walker moving fastest?

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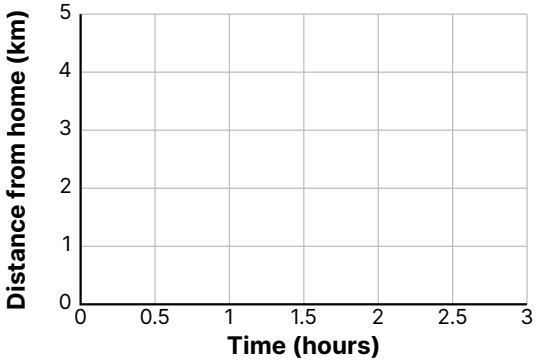
D Drawing a journey ●●●

Remember: Plot a point at the end of each part of the journey, then join the points with straight lines.

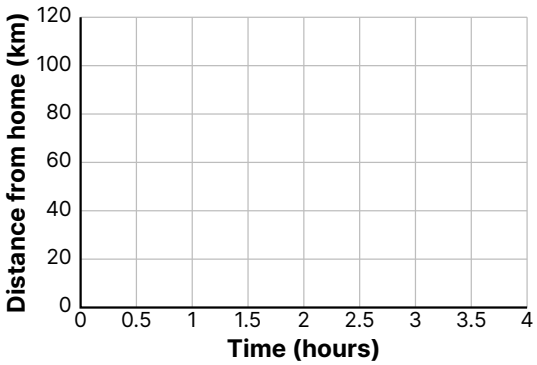
4. Aroha leaves home and cycles 6 km in 2 hours, then stops for 30 minutes, then cycles straight home, which takes 1 hour 30 minutes. Draw the journey on the grid.



5. Tane leaves home and walks 5 km in 1 hour, then stops for 30 minutes, then walks straight home, which takes 1 hour 30 minutes. Draw the journey on the grid.



6. A bus driver leaves home and drives 50 km in 1 hour, then stops for 1 hour, then drives 70 km further from home in 1 hour 30 minutes. Draw the journey on the grid.



Extension

7. Look again at the first graph on this sheet. Find the average speed for the whole ride, including the stop.

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